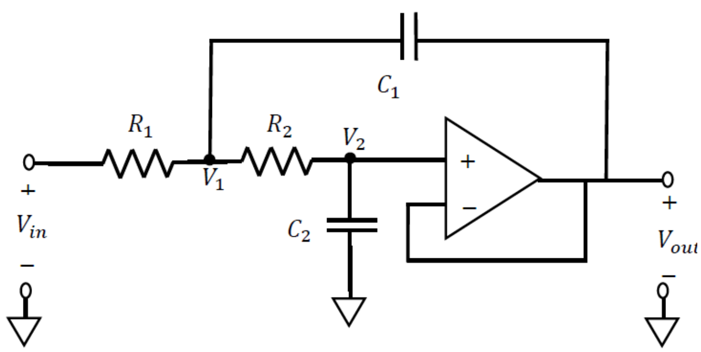
Lab 12 - Sinusoidal Steady-State Analysis

Objective

The goal of this lab is to further familiarize students with the frequency response of circuits to sinusoidal (AC) inputs. Students will connect the analysis techniques learned in class using complex phasors and frequency response to real world sinusoidal signals.

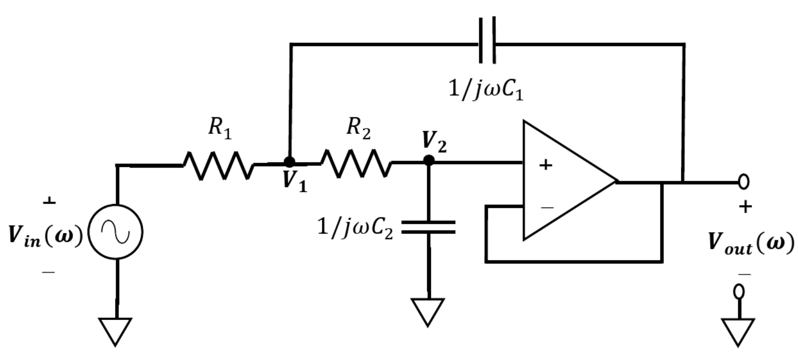
Introduction

As we continue to study 2nd order circuits, we will once again use the Sallen-Key circuit from an earlier lab (reproduced below). The key difference is that we will use an alternating current (AC) input to the circuit in this lab. While earlier we focused on the transient response of the circuit, now we will focus on the sinusoidal steady-state frequency response.



In class, you learned how to use phasors to find the steady state response of circuits with AC inputs. We will do the same with our Sallen-Key circuit. The phasor representation of the circuit is shown below.

Noting that $V_2 = V_{out}$ and performing KCLs at nodes 1 and 2 leads to the following equations:



$$\frac{V_{in}-V_1}{R_1}+\frac{V_{out}-V_1}{R_2}+\frac{V_{out}-V_1}{1/j\omega C_1}=0 \qquad \frac{V_1-V_{out}}{R_2}=\frac{V_{out}}{1/j\omega C_2}$$

Solving for V_1 in the second equation and substituting into the first (after a little algebraic manipulation) results in the following relationship between the input and output phasors:

$$V_{in}(\omega)=[1+j\omega(R_1+R_2)C_2-\omega^2 R_1 C_1 R_2 C_2]V_{out}(\omega)$$

It is common to describe the sinusoidal response of a circuit in terms of a quantity know as its *transfer function*, *H(ω)*, which is a function of frequency defined simply as the ratio of the output and input phasors, namely:

$$H(\omega)=\frac{V_{out}(\omega)}{V_{\text{in}}(\omega)}$$

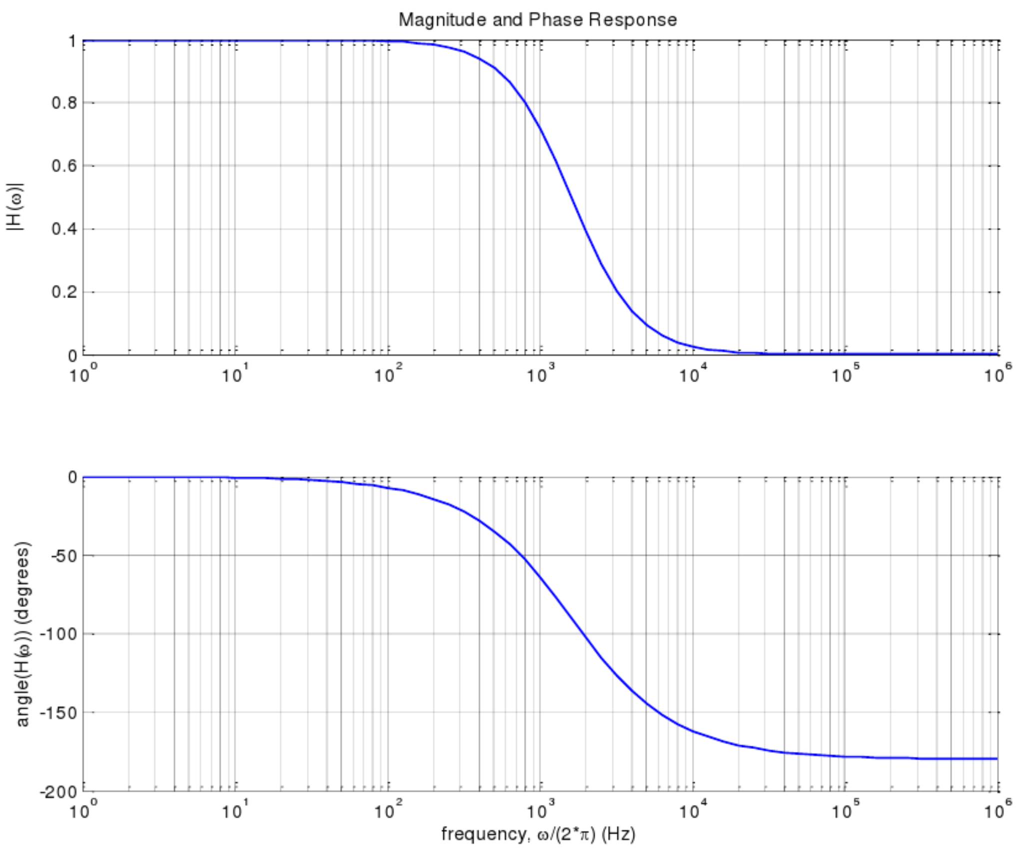
Given this definition and the relationship above, our Sallen-Key circuit has an H(ω) of:

$$H(\omega)=\frac{1}{1+j\omega(R_1+R_2)C_2-\omega^2 R_1 C_1 R_2 C_2}=\frac{1}{(1-\omega^2 R_1 C_1 R_2 C_2)+j\omega(R_1+R_2)C_2}$$

Converting this complex number into a magnitude and phase results in:

$$|H|=\frac{1}{\sqrt{(1-\omega^2 R_1 C_1 R_2 C_2)^2+(\omega(R_1+R_2)C_2)^2}} \qquad \phi=-\tan^{-1}\left(\frac{\omega(R_1+R_2)C_2}{1-\omega^2 R_1 C_1 R_2 C_2}\right)$$

Shown below are semi-log plots of H and ϕ as a function of frequency (ω/2π).



As seen from these graphs, as frequency goes to 0, H goes to 1 and as frequency goes to infinity, H goes to 0, thus this circuit is a *low-pass filter*, which we used in an earlier lab. This time, however, you can understand how and why it works.

Prelab

I. Low-Pass Filter

1. Design a Sallen-Key circuit as shown above. Choose component values (that are available in your lab kit) so that the circuit produces an underdamped response with Q=3/2 and $\omega_0 = 600\pi$ rad/sec ($f_0 = 300$Hz).
2. Simulate this circuit in LTSpice and copy the plot of the magnitude and phase response of your circuit similar to the ones shown. In your plot, the frequency should be in a logarithmic scale and should range from 10Hz to 10kHz.

II. High-Pass Filter

1. Modify your circuit by exchanging the positions of the resistors and the capacitors. *Keep the same component values as you used in part I*, just move their positions.
2. Develop an equation that provides the relationship between the phasor input and phasor output of the circuit (similar to the one above). From this relationship develop a formula for the magnitude and phase response of the circuit. Show your derivations.
3. Simulate this circuit in LTSpice and copy the plot of the magnitude and phase response of your modified circuit.

In-Lab Procedure

Parts and equipment needed:

- A selection of 1/4W resistors
- A selection of capacitors
- One 741 (or equivalent) op-amp
- A selection of colored 22 gauge connection wires
- XK700 Trainer
- Oscilloscope

It is a good idea to always verify your component values before you begin doing an experiment. This will allow you to identify one possible source of error easily.

I. Low-Pass Filter

1. Build the Sallen-Key circuit from the prelab using the component values you designed
2. Provide a sine wave as the input and display both input and output on your scope
3. Choose (at least) 15 frequencies between 100Hz and 10kHz for the input and measure the following for each frequency:
   - V_{in} and V_{out} peak-to-peak
   - $\Delta\phi$ between V_{out} and V_{in}
   - Note: For some frequencies, V_{out} is very small. In those cases, increase V_{in} so that V_{out} is a larger and easier to measure, if necessary
4. Adjust the frequency of the input until the amplitude ratio of V_{out}/V_{in} is 0.5
5. Record this (cut-off) frequency and its corresponding $\Delta\phi$

II. High-Pass Filter (Repeat the above)

1. Build the modified SK circuit from the prelab using the component values you designed
2. Provide a sine wave as the input and display both input and output on your scope
3. Choose (at least) 15 frequencies between 100Hz and 10kHz for the input and measure the following for each frequency:
   - V_{in} and V_{out} peak-to-peak
   - $\Delta\phi$ between V_{out} and V_{in}
   - Note: For some frequencies, V_{out} is very small. In those cases, increase V_{in} so that V_{out} is a larger and easier to measure, if necessary
4. Adjust the frequency of the input until the amplitude ratio of V_{out}/V_{in} is 0.5
5. Record this (cut-off) frequency and its corresponding $\Delta\phi$

Lab Report

- Title page
- Procedure: Write in your own words what you did in lab
- Derivations: Provide your derived equation for the magnitude and phase response for the modified Sallen-Key circuit (high-pass filter). You derivation should start from the relationship equation and end with a formula for the magnitude and phase response of the circuit (see Introduction for the approach used for the low-pass filter).
- Simulations
  - Include your LTSpice circuit diagrams and simulation plots.
- Data and Results:
  - Include your raw measured data from each task in a separate table
  - Provide a semi-log plot of the magnitude and phase responses for each of the circuits from your prelab design and your experimental results and compare the two.
  - Make sure to point out any significant differences you observed and try to explain the most likely causes of those differences.
  - Note: your measured responses should have 16 data points for each plot (the 15 frequencies chosen plus the cut-off frequency).
- Discussion:
  - Comment on the pass and stop bands of each circuit. That is, what frequencies pass through the circuit relatively unaltered and which frequencies are highly attenuated.
  - Given what you have learned about this circuit, what component values could you change (and how) to adjust the ranges of the pass bands and stop bands?
  - You should have observed that there are some frequencies where the output is stronger than the input. Discuss how that is even possible from a conservation of energy standpoint.
  - Be sure to point out any changes you could make to the procedure to make your results come out better if you had to do it all over again.